

Highlights

TSECAF-DETR: Representation-Query Co-Design for Real-Time Dense Wheat Head Detection

Zixin Wang, Jingchao Liu, Zizheng Zhao, Lifen Wen, Na Yang

- TSECAF-DETR frames dense detection as representation-query co-design.
- Token statistics guide semantic-detail exchange before proposal generation.
- Compact query allocation prioritizes scarce decoder capacity for small targets.
- Cross-dataset tests show stronger localization without target fine-tuning.

TSECAF-DETR: Representation-Query Co-Design for Real-Time Dense Wheat Head Detection

Zixin Wang^{a,*}, Jingchao Liu^a, Zizheng Zhao^a, Lifen Wen^a, Na Yang^a

^a*Xijing University, No. 1 Xijing Road, Chang'an District, Xi'an, Shaanxi Province, China*

Abstract

Dense small-object detection in cluttered imagery is constrained by a coupled representation-query bottleneck: compact targets must remain distinguishable after multi-scale aggregation and survive proposal-to-query competition before decoder refinement. Wheat head localization is a demanding case because heads are numerous, weakly separated, partially occluded, and visually similar to the surrounding canopy. This paper presents TSECAF-DETR, a real-time transformer detector that treats dense wheat head detection as a single evidence-to-query allocation problem rather than as separable feature-enhancement and query-selection tasks. The token-statistics-enhanced context-detail representation unit converts passive feature concatenation into statistics-guided semantic-detail exchange at the finest top-down fusion stage, forming compact candidate evidence before proposal generation. A matched compact query allocation rule combines objectness with an area-aware penalty, directing scarce decoder queries toward the same compact head-like candidates. The design preserves RT-DETRv2 end-to-end inference while aligning representation formation and decoder refinement capacity around dense small targets. Experiments on GWHD2021, the Wheat Ears Detection Dataset, and Global Wheat CodaLab evaluate in-domain detection, direct cross-dataset transfer, image-level count consistency, and inference speed. On GWHD2021, TSECAF-DETR-R50 achieved 53.62 AP, 93.18

*Corresponding author.

Email addresses: 2408540402005@stu.xijing.edu.cn (Zixin Wang), 20100058@xijing.edu.cn (Jingchao Liu), 2508540402019@stu.xijing.edu.cn (Zizheng Zhao), Wenlifen2023@163.com (Lifen Wen), 20140090@xijing.edu.cn (Na Yang)

AP50, 60.41 AP75, and 51.46 AP_S at 102.80 FPS. Without target-domain fine-tuning, it achieved 44.55 AP on the Wheat Ears Detection Dataset and 58.70 AP on Global Wheat CodaLab. It also reduced image-level counting MAE from 7.18 to 5.68 heads per image relative to RT-DETRv2-R50. These results indicate that representation-query co-design is an effective operating principle for dense small-object localization.

Keywords: dense small-object detection, wheat head localization, representation-query co-design, transformer detector, context-detail fusion, query allocation

1. Introduction

Detecting dense small objects in real-world images remains a persistent challenge for visual recognition systems. Small objects occupy few pixels, carry weak boundary evidence, and are easily suppressed by repeated down-sampling or coarse feature aggregation. The problem becomes more difficult when many instances are adjacent, partially occluded, and embedded in a background with similar local textures. In such scenes, detection quality is governed by a coupled bottleneck rather than by classification confidence alone: the representation must keep compact targets visually separable, and the decoder must allocate its limited refinement capacity to those targets before they disappear from the candidate set.

Wheat head localization is a representative dense small-object detection problem. Wheat heads are compact, numerous, weakly separated from leaves and awns, and frequently captured under variable illumination, viewpoint, resolution, and image quality. Public wheat-image benchmarks, especially the Global Wheat Head Detection dataset and its 2021 expansion, provide reproducible evaluation under diverse acquisition conditions [5, 6]. In this paper, wheat head detection is used as a challenging visual test bed for real-time dense small-object detection: source-domain accuracy measures localization quality, direct cross-dataset transfer measures robustness to acquisition shift, image-level count consistency measures whether detection errors accumulate in a simple downstream output, and throughput measures whether the detector remains practical for large image sets.

Figure 1 summarizes the accuracy-efficiency behavior observed in the source-domain comparison. TSECAF-DETR-R50 occupies the high-accuracy region among recent real-time detectors, achieving the best AP and AP_S

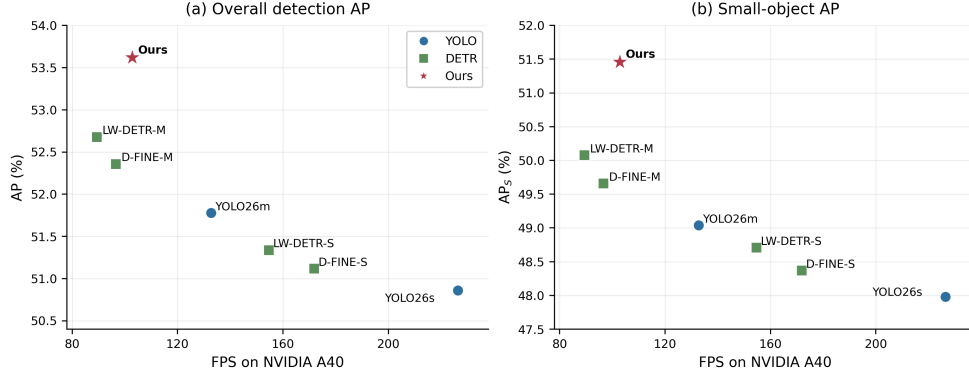


Figure 1: Accuracy-efficiency trade-off of recent real-time detectors on GWHD2021. TSECAF-DETR-R50 achieves the highest AP and AP_S among the compared models while maintaining real-time throughput on the NVIDIA A40 GPU, demonstrating a favorable balance between dense small-object localization accuracy and inference speed.

while retaining more than 100 FPS on the evaluation GPU. This operating point is important for dense image analysis because large image batches require both reliable localization and practical throughput for inspection, counting, and cross-domain comparison.

The performance advantage summarized in Figure 1 reflects a chained failure mode with two consecutive decisions. The high-resolution fusion stage determines whether weak local head evidence becomes a separable encoder candidate, while the proposal-to-query stage determines whether that candidate receives decoder refinement. If fusion blurs adjacent heads or suppresses weak boundaries, no compact hypothesis remains for the query selector to preserve. If query selection is driven only by objectness, compact hypotheses that were successfully formed can still be displaced by larger canopy-like proposals. Thus, representation formation and query allocation are not independent optimization targets; they operate on the same candidate population at successive points of the detector. Multi-scale CNN detectors and feature-pyramid variants address part of the first decision through scale-aware fusion [8–12]. Nevertheless, repeated downsampling and general-purpose fusion can create a semantic-detail mismatch: high-level semantics suppress ambiguity, while the local detail needed to separate neighboring small instances may be diluted. Transformer-based detectors provide end-to-end set prediction and global interaction, and RT-DETRv2 is attractive because it combines competitive accuracy with real-time inference potential [13–16]. However,

the default RT-DETRv2 design was developed for generic object detection. Its high-resolution fusion and confidence-driven query initialization are not explicitly organized around dense, weak-contrast compact objects, so valid candidates may either fail to form as distinct proposals or be filtered out before receiving decoder refinement.

To address these limitations, this paper develops TSECAF-DETR as a real-time transformer detector for dense wheat head detection. The key idea is to redesign one evidence-to-query pathway rather than to attach isolated accuracy-improving changes. Token-statistics-enhanced context-detail representation first strengthens the interaction between semantic context and local detail responses at the finest semantic-detail convergence point, where compact head evidence is most vulnerable to dilution. Compact-object query allocation then uses the resulting proposal geometry to increase the probability that those compact candidates are selected for decoder refinement. This forms a representation-query co-design in which the representation stage supplies compact hypotheses and the query stage protects them from being displaced by spatially larger responses. The detector’s internal information flow is therefore aligned with the geometry of dense small targets from high-resolution fusion to decoder initialization.

The main contributions of this article are summarized as follows:

1. A representation-query co-design framework is proposed for real-time dense wheat head detection. It treats high-resolution evidence formation and decoder query allocation as successive decisions over the same compact-candidate population rather than as independent detector details.
2. The Wheat-TS-ECAF representation unit converts high-resolution fusion into statistics-guided context-detail exchange. This mechanism uses global token statistics to regulate bidirectional semantic-detail transfer before proposal generation, making compact head hypotheses more explicit in cluttered images.
3. A compact-object query allocation strategy is formulated to align decoder initialization with the spatial scale of the hypotheses produced by the representation stage. The strategy ranks encoder proposals by joint objectness and area-aware scoring, increasing the decoder budget assigned to head-scale candidates without adding a separate post-processing stage.
4. A comprehensive evaluation is conducted through source-domain com-

parison, recent real-time detector benchmarking, direct cross-dataset transfer, image-level count consistency, pathway ablation, and hyperparameter sensitivity analysis. TSECAF-DETR-R50 achieves 53.62 AP and 51.46 AP_S on GWHD2021, reaches 44.55 AP on the Wheat Ears Detection Dataset without target-domain fine-tuning, and reduces GWHD2021 counting MAE from 7.18 to 5.68 heads per image relative to RT-DETRv2-R50.

The remainder of this article is organized as follows. Section 2 reviews related work on real-time object detection, wheat head detection, small-object feature fusion, and transformer-based query selection. Section 3 presents the TSECAF-DETR architecture. Section 4 describes the datasets, implementation details, evaluation metrics, and fairness protocol. Section 5 reports source-domain detection, recent real-time detector comparison, cross-dataset transfer, image-level counting, pathway ablation, and sensitivity results. Section 6 discusses the method behavior and current limitations of the study, and Section 7 concludes the article.

2. Related Work

2.1. Real-Time Object Detection in Dense Visual Scenes

Modern object detection has evolved from proposal-based localization to efficient one-stage and end-to-end formulations. Faster R-CNN remains a strong two-stage reference because it separates region proposal generation from final classification and regression [21]. SSD and RetinaNet improved efficiency by predicting objects directly from multi-scale features, with RetinaNet using focal loss to address foreground-background imbalance [22, 23]. EfficientDet further refined the accuracy-efficiency trade-off through weighted bidirectional feature fusion and compound scaling [10]. YOLO-family detectors are widely used when real-time inference is required, and recent versions such as Ultralytics YOLOv8, Ultralytics YOLO11, YOLOv12, and Ultralytics YOLO26 continue to improve speed and accuracy through stronger backbones, feature aggregation, and training recipes [24–28].

Dense small-object scenes remain difficult for these detector families. Compact targets may occupy only a small fraction of the input image, and their local evidence can be weakened by downsampling, feature fusion, or confidence ranking. When many targets are adjacent or visually similar to background structures, the detector must also use context to avoid false positives while preserving local details needed to separate neighboring instances.

These requirements expose a structural tension in detector design: the representation stage decides whether compact evidence remains separable, while the candidate allocation stage decides whether that evidence receives further refinement. This tension motivates designs that treat representation fusion and candidate allocation as coupled parts of the detection process rather than as independent implementation choices.

2.2. *Wheat Head Detection as Dense Small-Object Recognition*

Wheat head or spike detection has become a useful real-world benchmark for dense small-object recognition because heads are compact, repeated, partly occluded, and often embedded in cluttered plant textures. Early RGB-based studies used thresholding, hand-crafted features, or low-cost image-processing workflows under controlled or moderately complex conditions [18]. Deep learning improved robustness in more challenging images. Pound et al. introduced the ACID dataset and a deep multi-task architecture for spike and spikelet localization, showing that learned features can handle self-similarity, overlap, and viewpoint variation [31]. Hasan et al. released the SPIKE dataset and further demonstrated the usefulness of convolutional neural networks for wheat spike localization and counting [32].

Larger public benchmarks have made wheat head detection more suitable for reproducible visual evaluation. The Global Wheat Head Detection dataset and its 2021 expansion provide labeled images collected across countries, genotypes, platforms, and growing conditions [5, 6]. The Wheat Ears Detection Dataset further broadens image characteristics and annotation settings [33]. On these and related datasets, prior studies have investigated ear-density estimation from high-resolution RGB imagery [7], real-time YOLO-based wheat detectors [34], lightweight wheat head detection models [35, 36], augmented feature-pyramid counting pipelines [37], enhanced one-stage detectors with attention-guided feature fusion [38], counting-oriented methods such as DeepCount [39], semi-self-supervised learning under limited annotation [40], and transformer-based wheat head detection from UAV imagery [41].

These studies establish wheat head detection as a reproducible dense small-object benchmark with meaningful image-level outputs. They also indicate that source-domain AP alone is not sufficient. A detector trained on one dataset may face different image resolutions, illumination patterns, canopy densities, and annotation conventions in another dataset. For this

reason, this paper evaluates in-domain detection, cross-dataset transfer, image-level count consistency, and computational efficiency.

2.3. *Small and Dense Object Detection with Feature Fusion*

Small-object detection depends on whether the network can preserve fine spatial evidence while still using higher-level context. Feature Pyramid Networks introduced top-down fusion with lateral connections so that semantic information can be propagated to finer resolutions [8]. PANet added bottom-up path augmentation to shorten the information path between low-level localization cues and high-level features [9]. EfficientDet extended multi-scale fusion through the weighted bidirectional BiFPN design [10]. Scale-aware designs such as TridentNet and AugFPN further showed that object scale and pyramid-level imbalance can affect detection quality [11, 12]. Attention modules such as SE and CBAM demonstrated that channel and spatial recalibration can improve feature selectivity with limited overhead [42, 43].

These mechanisms provide useful foundations for dense wheat head detection, but wheat canopy imagery imposes a specific context-detail trade-off. Wheat heads are compact and often partly occluded, while leaves, stems, and awns form a repetitive background with similar local textures. Fine-resolution features are needed to separate adjacent heads and preserve boundary cues, whereas contextual features are needed to suppress non-head structures that resemble the target. If fusion is too coarse, neighboring heads may be merged or missed. If fusion is too local, the detector may confuse awns or leaves with true heads. Existing feature-pyramid and attention mechanisms do not explicitly target this task-specific tension, which motivates a context-detail representation process at the finest top-down stage. In this paper, that stage is treated as an information bottleneck: it must pass both discriminative context and instance-level detail to the proposal generator.

2.4. *Transformer-Based Detectors and Query Selection*

Transformer-based detection provides a complementary route by formulating object detection as end-to-end set prediction. DETR introduced this paradigm by replacing hand-crafted post-processing such as non-maximum suppression with learned object queries in a transformer architecture [13]. Deformable DETR improved convergence and multi-scale modeling by attending to sparse informative locations [14]. Later studies showed that query formulation and spatial priors strongly affect detector behavior. Conditional

DETR refined spatial query design for faster convergence [44]; SMCA introduced spatially modulated co-attention for location-aware decoding [45]; Anchor DETR assigned clearer spatial meaning to queries through anchor points [46]; and DAB-DETR represented queries as dynamic anchor boxes updated across decoder layers [47]. Sparse R-CNN, DN-DETR, and DINO further explored learnable proposals, denoising training, and improved anchor initialization [48–50]. More recent real-time transformer detectors, including D-FINE and LW-DETR, strengthen this direction through fine-grained localization refinement and lightweight transformer design [51, 52]. These works show that query initialization, localization refinement, and model efficiency are central to transformer detector performance.

RT-DETR adapts the DETR family to real-time detection through an efficient hybrid encoder and query initialization from encoder-side candidates [15]. RT-DETRv2 further improves this baseline while preserving the same design philosophy [16]. This detector family is attractive for dense-image analysis because it provides an end-to-end pipeline, scalable model variants, and practical inference efficiency. However, RT-DETRv2 was developed for generic detection benchmarks rather than dense wheat canopy imagery. When many valid targets are small, overlapping, and weakly separated from the background, the highest-resolution fusion stage and the allocation of limited decoder queries become especially important. Compact wheat head candidates with moderate encoder confidence may be underrepresented in the selected query set and therefore may not receive sufficient decoder refinement.

Most existing studies treat feature fusion and query selection as general detector subproblems. In dense wheat head detection, these two factors are coupled by the visual structure of the task and by the internal mechanics of proposal-based transformer detection. High-resolution fusion determines the quality and geometry of encoder proposals, while query selection decides which proposals become decoder inputs. A compact head that is merged during fusion cannot be recovered by a later query rule, and a compact proposal that is generated but not selected receives no decoder refinement. This gap motivates TSECAF-DETR, which formulates context-detail representation and compact-object query allocation as two linked decisions in one dense small-object detection pathway rather than as independent detector utilities. The resulting perspective turns wheat head detection from a crop-specific application into a focused case study of how transformer detectors allocate visual evidence and decoder capacity under dense small-object pressure.

3. Proposed Method

3.1. Overview of the TSECAF-DETR Architecture

TSECAF-DETR takes an RGB image as input and predicts bounding boxes with confidence scores for visible wheat heads. The resulting detections can be counted at the image level or inspected as localized object instances. Because wheat heads appear in dense canopy scenes rather than as isolated objects, the detector is organized to strengthen compact-head localization while maintaining real-time, end-to-end inference.

RT-DETRv2 [15, 16] is adopted as the baseline because it combines the end-to-end set-prediction formulation of DETR [13] with efficient multi-scale encoding and practical inference speed. Given an input image $\mathbf{I} \in \mathbb{R}^{3 \times H_0 \times W_0}$, the backbone extracts three stage-level feature maps with progressively reduced spatial resolution. After channel projection to a unified hidden dimension, the resulting feature set is denoted by $\{\mathbf{X}_3, \mathbf{X}_4, \mathbf{X}_5\}$, where $\mathbf{X}_3$ preserves the finest spatial information and $\mathbf{X}_5$ carries stronger semantic abstraction. In the baseline RT-DETRv2 pipeline, the highest-level feature is first processed by the transformer encoder, and the HybridEncoder then performs top-down and bottom-up aggregation to construct multi-scale features for encoder proposal generation and decoder refinement.

$$\{\mathbf{X}_3, \mathbf{X}_4, \mathbf{X}_5\} = \phi_{\text{proj}}(\phi_{\text{backbone}}(\mathbf{I})), \quad (1)$$

where $\phi_{\text{backbone}}(\cdot)$ denotes the convolutional backbone and $\phi_{\text{proj}}(\cdot)$ denotes the 1×1 channel projection before hybrid encoding. The encoded multi-scale representation predicts encoder-side candidate boxes and scores, written as $\{(s_i, \mathbf{b}_i)\}_{i=1}^{N_e}$. A fixed top- K subset initializes the decoder queries, and the decoder refines these candidates into the final detection set. This formulation is attractive for real-time dense detection because it supports end-to-end optimization and avoids hand-crafted post-processing such as non-maximum suppression [13, 15].

For dense wheat head imagery, the visual structure of the task creates two coupled design pressures. First, wheat heads are often characterized by weak boundaries, awn-like structures, and subtle local texture differences within a cluttered canopy. The semantic-detail convergence stage should therefore make canopy context and fine head-detail evidence interact explicitly, rather than simply aggregating them. Second, decoder initialization is a scarce-resource allocation problem: the fixed query budget should be assigned to candidate regions that are likely to correspond to individual compact heads

rather than broad canopy areas. These pressures are coupled because the query selector can only choose from the candidate evidence produced by the encoder representation, and the value of that representation depends on whether compact candidates survive into the decoder. This defines the central design question of TSECAF-DETR: how can a real-time transformer detector preserve compact visual evidence and then allocate decoder queries to that evidence?

TSECAF-DETR is built around this question. As shown in Figure 2, its representation process centers on Wheat-TS-ECAF, a token-statistics-enhanced context-detail representation unit placed at the finest semantic-detail convergence point of the hybrid encoder. This process produces a high-resolution representation in which semantic context and local boundary responses are exchanged before proposal generation, making compact head hypotheses more explicit in the encoder candidate set. The query allocation process then ranks those encoder proposals with a compact-object prior that combines objectness with predicted box area, guiding decoder initialization toward candidate heads that are most relevant to localization and count consistency. The two processes are deliberately coupled by proposal flow: the first shapes which compact candidates exist, and the second determines whether those candidates consume decoder queries. The detailed formulations of the representation and query allocation processes are given in Sections 3.2 and 3.3, respectively.

3.2. Token-Statistics-Enhanced Context-Detail Representation

In dense wheat canopies, missed and merged detections often occur when the detector fails to preserve fine head boundaries while using surrounding canopy context. The finest top-down fusion stage is important because it combines high-resolution local evidence with stronger semantic information before proposals are generated. In the baseline HybridEncoder, this stage receives an upsampled high-level feature and a lower-level high-resolution feature, concatenates them along the channel dimension, and forwards the result to the subsequent CSPRepLayer. Let $\mathbf{H}, \mathbf{L} \in \mathbb{R}^{C \times H \times W}$ denote the upsampled semantic feature and the projected low-level feature at this fusion stage, respectively. The baseline operation can be written as

$$\mathbf{F}_{\text{base}} = \text{Concat}(\mathbf{H}, \mathbf{L}), \quad (2)$$

where $\mathbf{F}_{\text{base}} \in \mathbb{R}^{2C \times H \times W}$ is then processed by the original FPN block. Although this design is computationally efficient, it treats the two branches as a

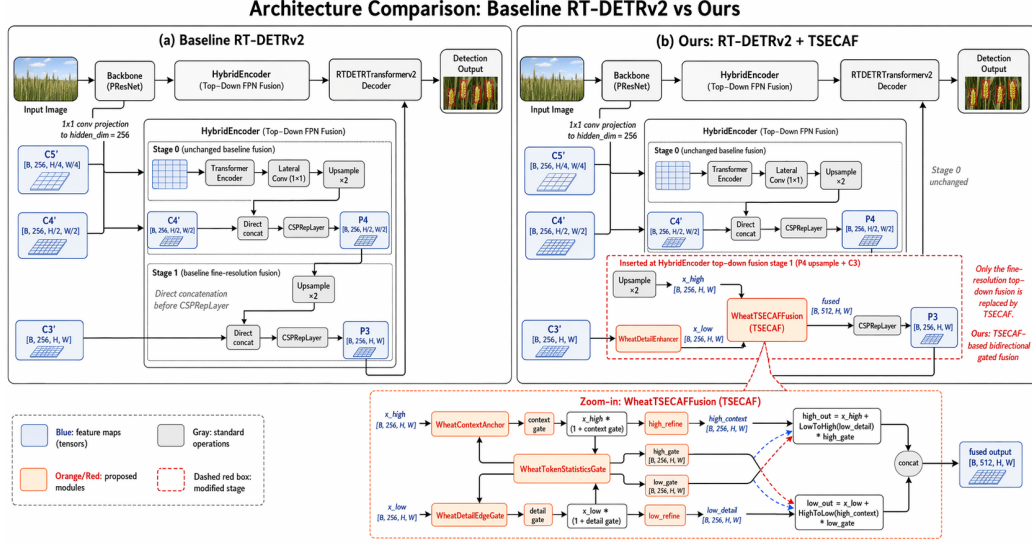


Figure 2: Architecture comparison between the baseline RT-DETRv2 and TSECAF-DETR for dense wheat head detection. TSECAF-DETR organizes the detector as a representation-query co-design over a shared proposal flow. The representation process strengthens the interaction between semantic context and local head-detail responses before proposal generation, while the query allocation process assigns decoder initialization to the compact candidates exposed by that representation.

passive aggregation pair and does not explicitly regulate how large-receptive-field semantics should interact with high-resolution detail cues. This limitation matters in wheat images, where a valid head may be defined by subtle edge responses and weak local contrast while still requiring contextual separation from nearby ears, leaves, and stems. It also affects the downstream query stage: if compact evidence is merged or suppressed at this point, later proposal ranking can only choose among already-degraded candidates.

TSECAF-DETR addresses this issue through Wheat-TS-ECAF, a token-statistics-enhanced context-and-detail-aware representation unit. Rather than treating high-level and low-level features as a passive concatenation pair, Wheat-TS-ECAF decomposes high-resolution fusion into context modeling, detail modeling, and statistics-guided bidirectional exchange. As shown in Figure 3, it contains a semantic/context branch, a detail/edge branch, and a global statistics gate that regulates information transfer between them. The LowToHigh and HighToLow transfer paths allow canopy-level semantics and

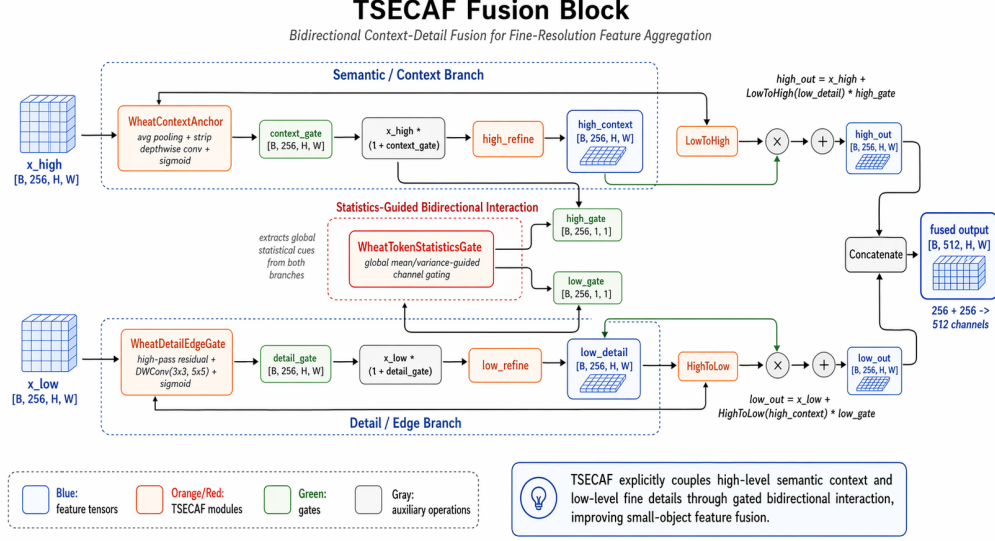


Figure 3: Structure of the Wheat-TS-ECAF representation unit. High-level semantic features and low-level detail features are processed through dedicated context and detail branches. WheatTokenStatisticsGate extracts branch-wise global statistics and drives gated bidirectional interaction through LowToHigh and HighToLow transfer paths, producing a context-detail representation tailored to compact heads in dense wheat canopies.

fine boundary responses to recalibrate one another before the representation is passed to proposal generation. In this sense, Wheat-TS-ECAF acts as a semantic-detail coordination layer: it does not merely enrich features, but adapts the contribution of each branch according to the global token distribution of the current image. Its role in the overall detector is therefore upstream and structural: it changes the candidate evidence that the compact-query policy will later allocate, rather than acting as an isolated feature enhancement block.

Context-aware high-level branch.. The upper branch in Figure 3 models region-level semantic context from the upsampled high-level feature. In wheat canopy images, local appearance alone is often insufficient because small ears may be partly occluded, visually blended with the canopy, or surrounded by plant structures with similar textures. A spatial context gate is therefore computed from $\mathbf{H}$ using a smoothed large-receptive-field transformation:

$$\mathbf{G}_c = \sigma(f_c(\mathcal{P}_7(\mathbf{H}))), \quad (3)$$

where $\mathcal{P}_7(\cdot)$ denotes 7×7 average pooling with stride 1, $f_c(\cdot)$ denotes the context-anchor mapping composed of channel reduction and two orthogonal depthwise strip convolutions, and $\sigma(\cdot)$ is the sigmoid function. The context-enhanced feature is then defined as

$$\mathbf{H}_c = \text{Refine}_h(\mathbf{H} \odot (1 + \mathbf{G}_c)), \quad (4)$$

where $\odot$ denotes element-wise multiplication and $\text{Refine}_h(\cdot)$ denotes a 3×3 depthwise refinement layer. Since $\mathbf{G}_c \in [0, 1]^{C \times H \times W}$, the multiplicative factor $(1 + \mathbf{G}_c)$ amplifies informative semantic regions without suppressing the original response to zero. This behavior is useful for wheat head detection because clustered ears often occupy contiguous canopy regions whose importance is better captured by contextual emphasis than by isolated pointwise activation.

Detail-aware low-level branch. The lower branch in Figure 3 focuses on local detail and boundary sensitivity. Small wheat heads are frequently separated from leaves, awns, and neighboring heads by narrow contour cues. Before entering Wheat-TS-ECAF, the finest projected feature is enhanced by a lightweight *WheatDetailEnhancer*, which is applied only to the lowest-level feature map:

$$\tilde{\mathbf{L}} = \delta(\mathbf{L}_0 + f_e([\text{DW}_3(f_p(\mathbf{L}_0)), \text{DW}_5(f_p(\mathbf{L}_0))])), \quad (5)$$

where $\mathbf{L}_0$ denotes the original projected low-level feature, $f_p(\cdot)$ is a 1×1 preprocessing layer, DW_3 and DW_5 are depthwise convolutions with kernel sizes 3 and 5, respectively, $f_e(\cdot)$ is the post-fusion projection, and $\delta(\cdot)$ is the nonlinear activation. For notational simplicity, the low-level input to the subsequent fusion block is still denoted by $\mathbf{L}$.

To emphasize fine wheat head contours, awn-like structures, and other high-frequency responses, the detail gate is generated from a high-pass residual:

$$\mathbf{R}_l = \mathbf{L} - \mathcal{P}_3(\mathbf{L}), \quad (6)$$

$$\mathbf{G}_d = \sigma(f_d([\text{DW}_3(\mathbf{R}_l), \text{DW}_5(\mathbf{R}_l)])), \quad (7)$$

where $\mathcal{P}_3(\cdot)$ denotes 3×3 average pooling with stride 1, and $f_d(\cdot)$ denotes the 1×1 fusion mapping used to generate the detail gate. The resulting detail-enhanced representation is

$$\mathbf{L}_d = \text{Refine}_l(\mathbf{L} \odot (1 + \mathbf{G}_d)), \quad (8)$$

where $\text{Refine}_l(\cdot)$ is a 3×3 depthwise refinement layer. Unlike the context branch, this process explicitly promotes spatially localized detail responses. It preserves edge evidence before semantic information is exchanged across branches, reducing the risk that adjacent heads are represented as a single merged region.

Statistics-guided bidirectional calibration and final fusion.. After obtaining $\mathbf{H}_c$ and $\mathbf{L}_d$, Wheat-TS-ECAF does not concatenate them directly. Instead, it estimates global branch-wise statistics to regulate information transfer in each direction. This avoids fixed fusion behavior across image scenes because images with severe canopy clutter and images with clearer head boundaries may require different context-detail balances. Spatial mean and variance are computed for both branches:

$$\begin{aligned}\boldsymbol{\mu}_h &= \text{Mean}(\mathbf{H}_c), \quad \boldsymbol{\nu}_h = \text{Var}(\mathbf{H}_c), \\ \boldsymbol{\mu}_l &= \text{Mean}(\mathbf{L}_d), \quad \boldsymbol{\nu}_l = \text{Var}(\mathbf{L}_d).\end{aligned}\tag{9}$$

where the statistics are computed over the spatial dimensions and lie in $\mathbb{R}^{C \times 1 \times 1}$. These four descriptors are concatenated and mapped by a 1×1 bottleneck network to produce two channel gates:

$$[\mathbf{G}_h, \mathbf{G}_l] = \text{Split}(\sigma(f_s([\boldsymbol{\mu}_h, \boldsymbol{\nu}_h, \boldsymbol{\mu}_l, \boldsymbol{\nu}_l]))),\tag{10}$$

where $\mathbf{G}_h, \mathbf{G}_l \in [0, 1]^{C \times 1 \times 1}$ are the statistics-guided channel gates for the two transfer directions, and $f_s(\cdot)$ denotes the bottleneck projection used in *WheatTokenStatisticsGate*. This design allows the representation unit to infer from the global response distribution whether semantic-to-detail transfer or detail-to-semantic transfer should be emphasized for a given input image.

The bidirectional interaction shown in Figure 3 is then written as

$$\mathbf{H}_{out} = \mathbf{H} + \psi_{lh}(\mathbf{L}_d) \odot \mathbf{G}_h,\tag{11}$$

$$\mathbf{L}_{out} = \mathbf{L} + \psi_{hl}(\mathbf{H}_c) \odot \mathbf{G}_l,\tag{12}$$

where $\psi_{lh}(\cdot)$ and $\psi_{hl}(\cdot)$ are 1×1 linear projections used to align transferred information before channel-wise modulation. The fused representation is then returned as

$$\mathbf{F}_{out} = \text{Concat}(\mathbf{H}_{out}, \mathbf{L}_{out}).\tag{13}$$

The final output $\mathbf{F}_{out} \in \mathbb{R}^{2C \times H \times W}$ has the same dimensional form as the baseline concatenation result in Equation (2), which keeps the representation

compatible with the downstream hybrid encoder while changing the interaction pattern at the semantic-detail convergence point. Overall, Wheat-TS-ECAF turns high-resolution fusion into an adaptive context-detail representation process in which contextual discrimination, boundary-sensitive detail, and statistics-guided exchange jointly define the compact evidence passed to proposal generation.

3.3. Compact-Object Query Allocation

In RT-DETRv2, decoder queries are initialized from top-ranked encoder proposals rather than learned independently from scratch. The quality of decoder initialization therefore depends directly on how encoder candidates are scored and selected. This design is important for dense wheat images because the number of possible head regions in a canopy can be much larger than the fixed decoder budget K . Query allocation is therefore the second half of the same evidence pathway shaped by high-resolution fusion: after compact candidates have been formed, they must still win a finite top- K competition. If this competition favors coarse or spatially large proposals, small but valid wheat heads may not enter decoder refinement, leading to missed detections that can bias counting or density estimation.

Let $\mathbf{m}_i \in \mathbb{R}^C$ denote the i -th encoder memory token after the encoder-output projection, where $i = 1, \dots, N_e$ and N_e is the total number of encoder candidates aggregated across all feature levels. In the TSECAF-DETR allocation process, the encoder score head outputs a single class-agnostic score logit

$$s_i = g_s(\mathbf{m}_i), \quad (14)$$

and the encoder box head outputs an unactivated box prediction

$$\mathbf{z}_i = g_b(\mathbf{m}_i) + \mathbf{r}_i, \quad (15)$$

where $g_s(\cdot)$ and $g_b(\cdot)$ denote the encoder score head and box head, respectively, and $\mathbf{r}_i$ is the anchor prior associated with the corresponding spatial location and pyramid level. The normalized proposal box is obtained with a sigmoid transform as

$$\hat{\mathbf{b}}_i = \sigma(\mathbf{z}_i) = (x_i, y_i, w_i, h_i), \quad (16)$$

from which the normalized proposal area is defined as

$$a_i = w_i h_i. \quad (17)$$

In the baseline class-agnostic strategy, proposals are ranked only by s_i . This is not ideal for dense wheat scenes because a high-confidence coarse proposal may correspond to a canopy region rather than an individual head. The proposal score therefore becomes an allocation policy for decoder capacity: candidates that win this ranking consume scarce decoder queries, whereas candidates that lose it receive no subsequent refinement. In TSECAF-DETR, this policy is matched to the upstream representation goal. Since Wheat-TSECAF is designed to make compact head evidence separable before proposal generation, the query ranking should not discard those compact proposals solely because larger regions produce high objectness. The proposed *agnostic_small* strategy therefore introduces a size-aware prior:

$$s_i^{\text{small}} = s_i - \lambda a_i, \quad (18)$$

where $\lambda \geq 0$ controls the strength of the small-object preference. The selected query indices are then obtained by

$$\mathcal{Q} = \text{TopK} \left(\{s_i^{\text{small}}\}_{i=1}^{N_e}, K \right), \quad (19)$$

and the corresponding encoder memory tokens, score logits, and proposal coordinates are gathered to initialize the decoder.

Algorithm 1 summarizes the compact-query allocation procedure. The ranking step is applied before decoder refinement; it determines which encoder candidates initialize decoder queries while leaving the decoder architecture and attention operations unchanged.

Equation (18) directly changes the ranking behavior. Under similar confidence scores, smaller proposals receive a weaker penalty and are more likely to survive top- K selection. This choice is suitable for single-class wheat head detection because the ranking objective does not need to resolve competition among multiple semantic categories; instead, it should retain candidate regions that are useful for refining individual head locations. As illustrated in Figure 4, the standard path performs top- K selection using objectness alone, whereas the TSECAF-DETR compact-query path first derives the predicted box, computes its area, and then forms a size-aware score before selection. This allocation design introduces no additional post-processing and preserves end-to-end inference, but it changes the detector’s internal policy: decoder queries are treated as a limited resource assigned to the compact instance hypotheses that the representation stage is designed to expose.

Algorithm 1 Small-object-aware query selection for dense wheat head detection

Input: encoder memory tokens $\{\mathbf{m}_i\}_{i=1}^{N_e}$, anchor priors $\{\mathbf{r}_i\}_{i=1}^{N_e}$, decoder budget K , area penalty coefficient λ

Output: selected query embeddings, proposal boxes, and scores for decoder initialization

1. Predict the class-agnostic score logit $s_i = g_s(\mathbf{m}_i)$ for each encoder candidate.
 2. Predict the proposal logit $\mathbf{z}_i = g_b(\mathbf{m}_i) + \mathbf{r}_i$ and obtain the normalized proposal box $\hat{\mathbf{b}}_i = \sigma(\mathbf{z}_i) = (x_i, y_i, w_i, h_i)$.
 3. Compute the normalized proposal area $a_i = w_i h_i$.
 4. Form the small-object-aware ranking score $s_i^{\text{small}} = s_i - \lambda a_i$.
 5. Select $\mathcal{Q} = \text{TopK}(\{s_i^{\text{small}}\}_{i=1}^{N_e}, K)$.
 6. Gather $\{\mathbf{m}_i, \hat{\mathbf{b}}_i, s_i\}_{i \in \mathcal{Q}}$ to initialize decoder queries and reference boxes.
-

Anchor prior adjustment. The size-aware ranking in Equation (18) is complemented by a smaller anchor prior during proposal generation. This adjustment reflects the fact that wheat heads occupy relatively compact image regions compared with many generic detection targets. For each pyramid level ℓ with spatial size $H_\ell \times W_\ell$, the anchor center at grid location (u, v) is defined as

$$\mathbf{c}_{u,v,\ell} = \left(\frac{u + 0.5}{W_\ell}, \frac{v + 0.5}{H_\ell} \right), \quad (20)$$

and the corresponding normalized anchor width and height are initialized as

$$\mathbf{d}_\ell = (g \cdot 2^\ell, g \cdot 2^\ell), \quad (21)$$

where g is the anchor-grid-size hyperparameter. The anchor prior before inverse-sigmoid transformation is written as

$$\bar{\mathbf{r}}_{u,v,\ell} = [\mathbf{c}_{u,v,\ell}, \mathbf{d}_\ell]. \quad (22)$$

In implementation, valid anchors are transformed to the logit domain and added to the encoder box regression output, as shown in Equation (15). Reducing g shifts the initial proposal prior toward smaller boxes at every feature level, making compact wheat-head hypotheses easier to form before ranking.

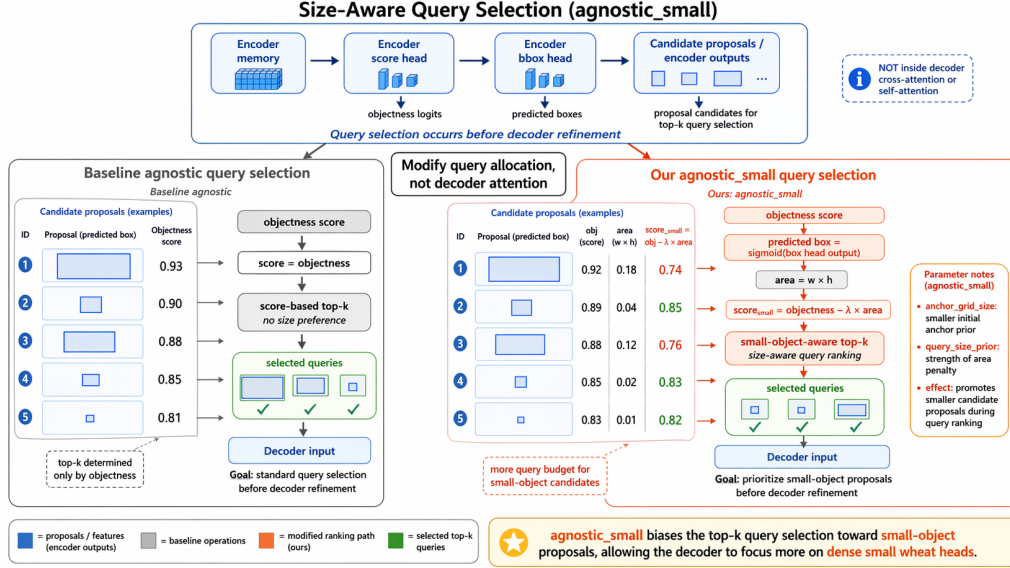


Figure 4: Comparison between the baseline class-agnostic query selection and the TSECAF-DETR *agnostic_small* allocation rule for dense wheat head detection. Instead of ranking encoder proposals only by objectness, TSECAF-DETR computes a size-aware score, $s^{\text{small}} = \text{objectness} - \lambda \times \text{area}$, so that compact candidates are more likely to be retained during top- K query selection. This makes decoder initialization consistent with the compact evidence emphasized by the representation pathway in crowded canopy regions.

The two compact-query mechanisms operate at different stages of the same allocation process. The reduced anchor prior defines the geometric starting point of encoder proposals, whereas the *agnostic_small* score determines which proposals consume decoder queries. In the final configuration used in this article, $K = 500$, $g = 0.04$, and $\lambda = 8.0$ jointly shape the decoder budget around compact wheat-head candidates. This coupling between proposal prior and query ranking is particularly relevant to dense wheat scenes because it increases the likelihood that small but valid head instances enter decoder refinement instead of being dominated by spatially larger canopy candidates.

4. Experimental Setup

The experimental protocol was designed to evaluate TSECAF-DETR as a dense small-object detector rather than only as a source-domain model. The

protocol was organized around four questions. First, the model should improve source-domain localization compared with representative CNN, YOLO, and RT-DETRv2 baselines. Second, the improvement should remain observable when a GWHD2021-trained checkpoint is evaluated directly on external wheat datasets, because changes in image source, resolution, annotation style, and canopy structure create realistic domain shifts. Third, the localization gain should reduce image-level counting error, because count consistency is a simple way to measure whether missed and duplicate detections accumulate in dense scenes. Fourth, the observed improvement should be attributable to the coordinated context-detail representation and compact-query allocation processes rather than to an incidental configuration choice. This section describes the datasets, implementation details, evaluation metrics, and fairness controls used in the experiments.

4.1. Datasets and Evaluation Protocol

GWHD2021 was used as the source dataset for model training and in-domain evaluation [6]. It contains 6515 images with diverse wheat-head appearances, cultivar backgrounds, image qualities, and head densities. For all source-domain experiments, the same fixed split was used, with 5499 images for training and 1016 images for validation. To evaluate behavior beyond the training distribution, GWHD2021-trained checkpoints were also tested on two external wheat datasets: the Wheat Ears Detection Dataset [33] and the labeled subsets of the official Global Wheat CodaLab release [53]. No target-domain fine-tuning was performed in these transfer experiments. The reported target-domain scores therefore measure direct cross-dataset transfer rather than retuning the detector for each target dataset.

All auxiliary datasets were converted to the COCO-style layout used by the RT-DETRv2, Faster R-CNN, and RetinaNet pipelines [54]. For the Wheat Ears Detection Dataset, the original XML annotations were parsed into COCO bounding-box records. Its 236 high-resolution images with 30,729 ear annotations were divided into train2017 and val2017 partitions using a fixed 80:20 split. For the Global Wheat CodaLab release, only labeled subsets were used for quantitative evaluation; each subset was split independently with the same 80:20 ratio to preserve domain diversity during validation. In the transfer experiments, these target-domain training partitions were created only for dataset-format consistency and were not used for model fitting. Quantitative transfer scores were computed on the corresponding validation partitions. Subset-name prefixes were added to filenames to avoid collisions

after images from different domains were merged into unified folders. For YOLO-family detectors, the resulting COCO annotations were further converted into Ultralytics label files by the project scripts so that all models used matched image and annotation sets.

4.2. Implementation Details and Fair Comparison Protocol

All experiments were conducted on a single NVIDIA A40 GPU using the project implementations. Unless otherwise specified, the input resolution was fixed at 1024×1024 for training-time validation, source-domain evaluation, and efficiency measurement so that accuracy and speed were measured at the same image scale. The training batch size was 8 images for the RT-DETRv2 and TSECAF-DETR variants, Faster R-CNN, RetinaNet, YOLO-family baselines, D-FINE baselines, and LW-DETR baselines; all efficiency measurements used a batch size of 1. All models were initialized from the official open-source pretrained weights released by their respective implementations before training on the GWHD2021 source split [16, 25, 26, 28, 51, 52, 55]. To keep the paired comparison focused on the TSECAF-DETR architecture, each RT-DETRv2 baseline and its corresponding TSECAF-DETR variant used the same backbone family, source dataset, training schedule, and evaluation protocol. For RT-DETRv2 and TSECAF-DETR, the optimizer was AdamW [56] with a base learning rate of 5×10^{-5} , a backbone learning rate of 5×10^{-6} , and a weight decay of 10^{-4} . The R18 model was trained for 120 epochs, whereas the R34 and R50 variants were trained for 180 epochs. The Faster R-CNN, RetinaNet, YOLOv8, YOLO11, YOLO26, D-FINE, and LW-DETR baselines were trained for 180 epochs on the same GWHD2021 training split. Multi-scale training for the RT-DETRv2 and TSECAF-DETR variants used resize scales from 768 to 1088 pixels. On GWHD2021, data augmentation included random photometric distortion, random zoom-out, random IoU crop, random horizontal flipping, and multi-scale resizing, with stronger geometric augmentations disabled during the final training stage. The checkpoint with the best GWHD2021 validation performance was used for all transfer evaluations.

TSECAF-DETR instantiates its context-detail representation process at the highest-resolution semantic-detail convergence stage of the HybridEncoder. In the final configuration, compact-object query allocation used 500 object queries, an anchor grid size of 0.04, and a size-prior weight of 8.0. Following the repository settings, the ResNet-based R18, R34, and R50 variants [57] used 3, 4, and 6 decoder layers, respectively, with 100, 150, and 200

denoising queries. This setup focuses the comparison on dense small-object representation and query allocation while keeping backbone scale, training schedule, and inference protocol controlled.

As source-domain reference points, Faster R-CNN-R50-FPN [21], RetinaNet-R50 [23], Ultralytics YOLOv8s and YOLOv8m [25], and Ultralytics YOLO11s and YOLO11m [26] were trained on the same GWHD2021 split. Additional recent real-time baselines, including YOLO26s/m, D-FINE-S/M, and LW-DETR-S/M, were evaluated under the same source-domain split, input scale, and GPU measurement setting to compare TSECAF-DETR with newer YOLO and DETR-family models. The YOLO-family models used AdamW optimization, cosine learning-rate decay [58], and mosaic/mixup augmentation [24, 59]. The TorchVision-style Faster R-CNN and RetinaNet implementations used AdamW, cosine annealing, large-scale jitter, reduced anchor sizes, and GIoU box regression [55, 60]. Detection metrics were computed on the fixed GWHD2021 validation split. Parameters, GFLOPs, and FPS were measured for the final selected checkpoint of each model. FPS was measured on the same A40 GPU with batch size 1 and a 1024×1024 input after 100 warm-up iterations, then repeated five times over the 1016-image validation set; visualization and disk I/O were excluded. Standard NMS-based post-processing was included for Faster R-CNN, RetinaNet, YOLOv8, and YOLO11 detectors, whereas YOLO26, RT-DETRv2, and TSECAF-DETR variants used their native end-to-end outputs without NMS.

For fairness, all models in the main source-domain comparison used the same training and validation images, the same 1024×1024 evaluation scale, the same COCO-style evaluation script, and the same GPU environment for efficiency measurement. Because different detector families rely on family-specific standard training recipes, fairness was enforced under the same dataset, input scale, epoch budget, and evaluation protocol within each comparison group, while each model family retained its standard optimization and augmentation settings. This setting reflects a practical detector comparison rather than forcing all architectures into an identical hyperparameter recipe that may disadvantage some model families. For precision, recall, F1, and image-level counting, each model used the validation confidence threshold that maximized F1. This avoided manually selecting a favorable threshold for any specific detector and ensured that the derived counting comparison used a consistent operating point.

4.3. Evaluation Metrics and Counting Protocol

Detection accuracy was primarily evaluated using COCO-style AP@[0.50:0.95] [54], abbreviated as AP, which averages detection quality across IoU thresholds from 0.50 to 0.95. AP50 reflects localization under a relaxed overlap threshold, AP75 indicates stricter localization quality, and AP_S measures small-object behavior. Precision, recall, and F1 were computed after confidence filtering at the validation threshold that maximized F1 for each model. For the Wheat Ears Detection Dataset, AR300 was reported because dense scenes in this dataset can contain substantially more than 100 wheat heads per image; under such conditions, the default COCO detection limit may underestimate recall. The maximum detections for Wheat Ears evaluation were therefore set to [1, 10, 300] in all frameworks. The number of trainable parameters, GFLOPs, and inference speed in frames per second (FPS) were reported as practical indicators of computational cost.

To examine whether localization improvements translated into a useful image-level output, wheat-head counting was evaluated on the 1016-image GWHD2021 validation set. No separate counting model was trained. For each image, the ground-truth count was defined as the number of annotated wheat-head boxes, and the predicted count was obtained by counting the final predicted boxes after confidence filtering. This evaluates whether detection errors accumulate when localized instances are converted into a simple count. Counting accuracy was measured using mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), and the coefficient of determination (R^2).

5. Experimental Results and Analysis

This section reports the empirical results obtained under the protocol described in Section 4. The analysis is organized around five aspects: source-domain detection on GWHD2021, comparison with recent real-time detectors, direct transfer to external wheat datasets, image-level counting behavior, and pathway-level ablation.

5.1. Source-Domain Detection Performance on GWHD2021

Table 1 compares source-domain detection accuracy and computational efficiency on GWHD2021 against representative two-stage, one-stage, YOLO, and RT-DETRv2 baselines. This setting evaluates TSECAF-DETR not only

Table 1: Source-domain comparison on GWHD2021. Detection metrics were computed on the fixed validation split. FPS was measured with a batch size of 1 on an NVIDIA A40 GPU.

Model	Input	AP	AP50	AP75	AP-S	Precision	Recall	F1	Params	GFLOPs	FPS
Faster R-CNN-R50-FPN	1024	47.82	89.44	53.11	44.73	88.12	82.36	85.14	41.3M	182.6	42.7
RetinaNet-R50	1024	49.18	90.86	54.92	46.31	88.74	83.95	86.28	36.4M	161.8	79.6
YOLOv8s	1024	48.97	90.31	54.48	45.96	89.65	83.22	86.31	11.1M	71.4	226.5
YOLOv8m	1024	50.42	91.37	56.31	47.52	90.28	84.91	87.51	25.9M	189.2	132.4
YOLO11s	1024	49.76	91.02	55.71	46.83	90.11	84.18	87.04	9.4M	66.8	238.6
YOLO11m	1024	50.71	91.58	56.84	47.91	90.63	85.24	87.85	20.1M	182.5	145.2
RT-DETRv2-R18	1024	48.76	89.94	54.06	45.88	88.91	83.47	86.10	21.3M	86.5	213.7
TSECAF-DETR-R18	1024	50.28	91.08	56.07	48.31	90.16	85.02	87.51	21.9M	89.7	207.4
RT-DETRv2-R34	1024	50.03	91.27	55.88	47.62	90.04	84.68	87.28	31.3M	128.4	158.7
TSECAF-DETR-R34	1024	51.62	91.96	58.11	49.45	91.02	86.11	88.50	31.9M	132.1	154.2
RT-DETRv2-R50	1024	50.68	91.74	56.92	47.96	90.51	85.23	87.79	42.1M	171.6	108.5
TSECAF-DETR-R50	1024	53.62	93.18	60.41	51.46	92.14	87.61	89.82	42.8M	176.3	102.8

against its paired RT-DETRv2 baseline, but also against commonly used detector families for dense object detection, where detector choice affects both localization accuracy and the time required to process large image collections.

The TSECAF-DETR variants improved all three RT-DETRv2 backbone scales. Relative to the paired baselines, AP increased by 1.52 points for R18, 1.59 points for R34, and 2.94 points for R50. The corresponding AP_S gains were 2.43, 1.83, and 3.50 points, respectively. The larger gains on AP_S indicate that context-detail representation and compact-query allocation primarily benefit compact wheat heads, which is the intended operating regime of the detector. Among the standard baselines in Table 1, TSECAF-DETR-R50 achieved the highest accuracy, with 53.62 AP, 93.18 AP50, 60.41 AP75, and 51.46 AP_S . Compared with YOLO11m, the strongest YOLO baseline in this comparison, TSECAF-DETR-R50 improved AP by 2.91 points and AP_S by 3.55 points. YOLO-family models remained faster and lighter, especially YOLO11s and YOLOv8s, but TSECAF-DETR-R50 provided stronger localization accuracy for compact heads while maintaining real-time inference at 102.80 FPS.

The three TSECAF-DETR model scales also provide different operating points. TSECAF-DETR-R18 reached 207.36 FPS with 50.28 AP, TSECAF-DETR-R34 provided an intermediate balance, and TSECAF-DETR-R50 achieved the highest accuracy. This range is useful because some dense-image workloads prioritize rapid batch inference, whereas others prioritize stricter localization and lower count error.

Table 2 reports the repeated FPS measurement protocol for the final TSECAF-DETR-R50 model. The five repeated runs produced a mean throughput of 102.80 FPS on the NVIDIA A40 GPU. The F1-max threshold used

Table 2: Repeated FPS measurement protocol for the final TSECAF-DETR-R50 model on GWHD2021.

GPU	Input	Batch	Warm-up	Timed images	FPS runs	Mean FPS
NVIDIA A40	1024	1	100	1016×5	102.42/102.76/102.91/103.05/102.86	102.80

Table 3: Additional source-domain comparison with recent real-time detectors on GWHD2021. All models were trained for 180 epochs at 1024×1024 resolution with a batch size of 8 and evaluated on the same validation split. FPS was measured on an NVIDIA A40 GPU with batch size 1 after warm-up. Delta AP is computed relative to TSECAF-DETR-R50.

Model	AP	AP50	AP75	AP-S	Precision	Recall	F1	Params	GFLOPs	FPS	Epoch	Batch	Input	Delta AP
YOLO26s	50.86	91.34	56.42	47.98	90.72	84.94	87.73	10.2M	72.9	226.40	180	8	1024	-2.76
YOLO26m	51.78	92.03	57.86	49.04	91.18	85.92	88.47	22.4M	190.6	132.70	180	8	1024	-1.84
D-FINE-S	51.12	91.66	56.95	48.37	90.88	85.21	87.95	19.3M	82.5	171.80	180	8	1024	-2.50
D-FINE-M	52.36	92.31	59.02	49.66	91.46	86.42	88.87	31.6M	156.4	96.50	180	8	1024	-1.26
LW-DETR-S	51.34	91.82	57.20	48.71	91.02	85.47	88.16	15.8M	76.2	154.60	180	8	1024	-2.28
LW-DETR-M	52.68	92.57	59.38	50.08	91.72	86.66	89.12	28.7M	147.9	89.30	180	8	1024	-0.94
TSECAF-DETR-R50	53.62	93.18	60.41	51.46	92.14	87.61	89.82	42.8M	176.3	102.80	180	8	1024	0.00

for precision, recall, F1, and image-level counting was 0.286.

5.2. Comparison With Recent Real-Time Detectors

Table 3 further compares TSECAF-DETR-R50 with recent real-time YOLO and DETR-family detectors under the same source-domain protocol. YOLO26m improved over earlier YOLO baselines, but it remained 1.84 AP and 2.42 AP_S below TSECAF-DETR-R50. TSECAF-DETR-R50 also outperformed D-FINE-M and LW-DETR-M by 1.26 and 0.94 AP, respectively, while running 6.30 and 13.50 FPS faster than these two medium-size transformer baselines. The advantage was particularly clear for AP_S, where TSECAF-DETR-R50 exceeded D-FINE-M and LW-DETR-M by 1.80 and 1.38 points, respectively. These results indicate that TSECAF-DETR provides a favorable small-object accuracy and speed trade-off for dense wheat head detection.

Additional inference results produced by TSECAF-DETR-R50 on the GWHD2021 validation set are shown in Figure 5. These examples demonstrate the detector’s behavior in dense wheat canopies: compact and partially overlapping heads are separated with stable confidence, and detections remain consistent under low contrast, image blur, and high target density. This qualitative evidence is consistent with the higher quantitative scores reported in Tables 1 and 3, especially the improvements in AP75 and AP_S, by showing that TSECAF-DETR-R50 preserves head-level localization in scenes where missed or merged boxes would directly degrade counting accuracy.

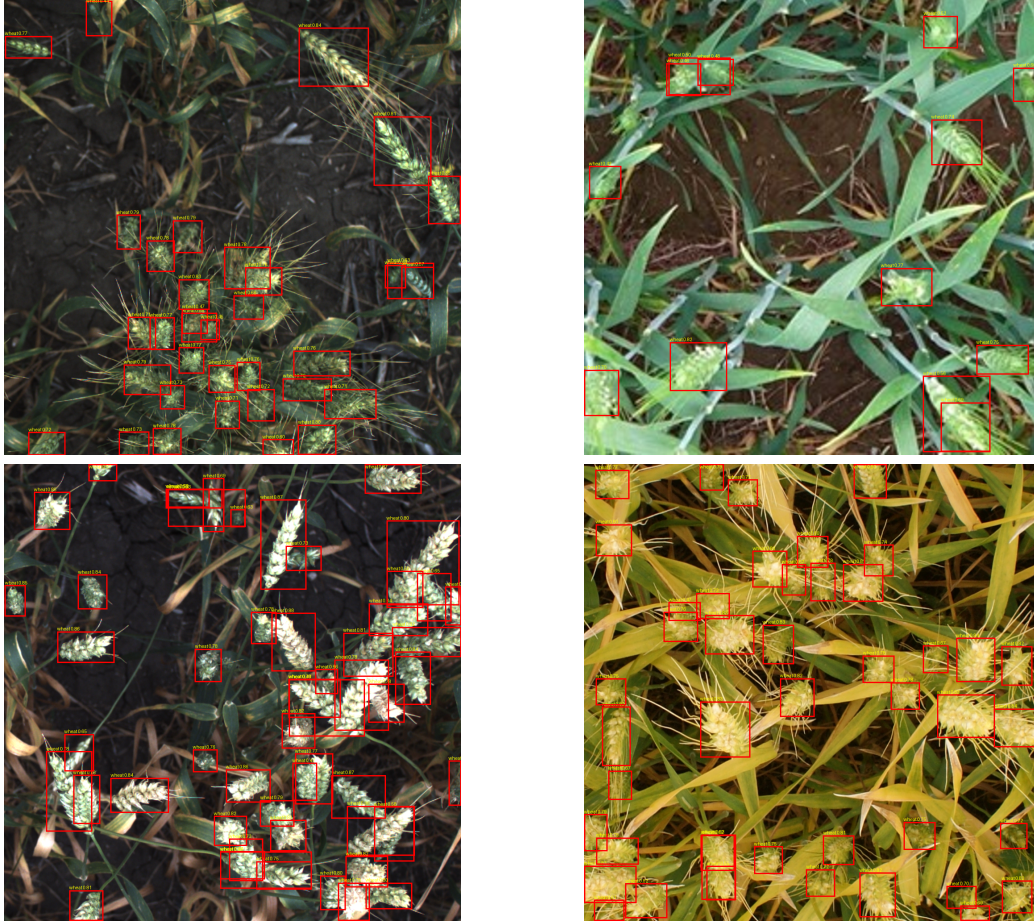


Figure 5: Additional wheat head detection results produced by TSECAF-DETR-R50 on GWHD2021 validation images. The detections demonstrate robust head-level localization under low contrast, image blur, and dense overlapping targets, supporting the model’s higher AP, AP_S, and counting performance. Red boxes denote predicted wheat head detections.

5.3. Cross-Dataset Transfer Performance

Cross-dataset transfer is important because images acquired from different sources may differ in spike density, canopy clutter, illumination, viewing height, camera properties, and annotation style. This setting reflects the broader transfer-learning problem in which training and target distributions do not fully match [61, 62]. In all transfer experiments, the detector was trained on GWHD2021 and then evaluated directly on the target valida-

Table 4: Direct transfer from GWHD2021 to external wheat datasets without target-domain fine-tuning.

Model	Wheat Ears AP	Wheat Ears AP50	Wheat Ears AR300	CodaLab AP	CodaLab AP50
RetinaNet-R50	36.82	84.91	47.36	53.74	93.62
YOLOv8s	37.94	85.88	48.07	54.38	93.95
YOLOv8m	39.26	86.73	49.42	55.41	94.32
YOLO11s	38.71	86.24	48.85	55.02	94.18
YOLO11m	40.02	87.31	50.05	55.76	94.51
RT-DETRv2-R50	40.36	87.92	50.18	56.09	94.88
TSECAF-DETR-R50	44.55	90.28	54.62	58.70	95.94

tion partition without target-domain fine-tuning. The goal was not to claim universal robustness, but to test whether the detector remains useful when moved from one public image source to another.

Table 4 summarizes direct transfer to the Wheat Ears Detection Dataset and the labeled subsets of Global Wheat CodaLab. The Wheat Ears Detection Dataset provides a demanding transfer test because its high-resolution images contain very dense ear distributions and place strong pressure on small-object recall. Under this shift, TSECAF-DETR-R50 achieved 44.55 AP, 90.28 AP50, and 54.62 AR300, outperforming the paired RT-DETRv2-R50 baseline by 4.19 AP and 4.44 AR300. It also exceeded YOLO11m, the strongest YOLO-family transfer baseline, by 4.53 AP. This behavior is consistent with the intended role of TSECAF-DETR: context-detail representation preserves fine spatial evidence, while compact-query allocation assigns more decoder capacity to head-scale candidates.

The labeled subsets of Global Wheat CodaLab showed a milder but consistent trend. TSECAF-DETR-R50 reached 58.70 AP and 95.94 AP50, improving over the paired RT-DETRv2-R50 baseline by 2.61 AP and over YOLO11m by 2.94 AP. The gain was smaller than that on Wheat Ears, which is reasonable because the Global Wheat CodaLab evaluation represents a broader but less uniformly dense domain shift. Together, these transfer results indicate that TSECAF-DETR improves cross-dataset generalization rather than simply fitting the source-domain validation split.

5.4. Image-Level Counting Performance

To examine whether improved localization translates into a useful image-level output, wheat-head counts were estimated by counting the final predicted boxes for each image. Table 5 reports image-level counting errors on the GWHD2021 validation set. TSECAF-DETR-R50 achieved the lowest MAE, RMSE, and MAPE and the highest R^2 . Compared with the paired

Table 5: Image-level wheat-head counting performance on the 1016-image GWHD2021 validation set. Predicted counts were obtained by counting confidence-filtered detection boxes at the F1-max threshold of each model.

Model	F1-thr	MAE (lower)	RMSE (lower)	MAPE (lower)	R-squared (higher)
RetinaNet-R50	0.302	8.74	12.63	13.82%	0.872
YOLOv8s	0.287	8.31	12.07	13.16%	0.884
YOLOv8m	0.279	7.62	11.34	12.08%	0.901
YOLO11s	0.284	7.94	11.68	12.51%	0.893
YOLO11m	0.276	7.41	11.02	11.74%	0.907
YOLO26m	0.274	6.86	10.15	10.82%	0.925
D-FINE-M	0.261	6.47	9.69	10.22%	0.934
LW-DETR-M	0.268	6.21	9.38	9.83%	0.939
RT-DETRv2-R50	0.281	7.18	10.81	11.39%	0.913
TSECAF-DETR-R50	0.286	5.68	8.71	8.94%	0.948

RT-DETRv2-R50 baseline, MAE decreased from 7.18 to 5.68 heads per image, RMSE decreased from 10.81 to 8.71, and R^2 increased from 0.913 to 0.948. Compared with LW-DETR-M, the strongest recent counting baseline in this experiment, TSECAF-DETR-R50 reduced MAE by 0.53 heads per image while also improving count-level correlation. These results indicate that the detection improvement is not limited to AP metrics; it also reduces the error of a direct image-level count derived from detection boxes.

Because the proposed design targets dense small-object scenes, the counting results should be interpreted together with the detection metrics most related to compact targets. The simultaneous improvement in AP75, AP_S, and image-level MAE, summarized in Figure 6, suggests that better localization of compact and partially overlapping heads can reduce downstream counting error.

5.5. Pathway Ablation and Sensitivity Analysis

The ablation study examined how controlled probes of the TSECAF-DETR evidence-to-query pathway affect the final detector. RT-DETRv2-R50 was used as the controlled reference model on GWHD2021. Table 6 evaluates three controlled interventions within this formulation: context-detail representation, detail-branch emphasis, and compact-object query allocation. These interventions should be read as pathway tests rather than as separable upgrades, because each changes either the compact evidence available to proposal heads or the allocation of decoder capacity to that evidence. All variants were trained under the same source-domain setting as the main experiment. In addition to AP and AP50, AP75 and AP_S are reported because

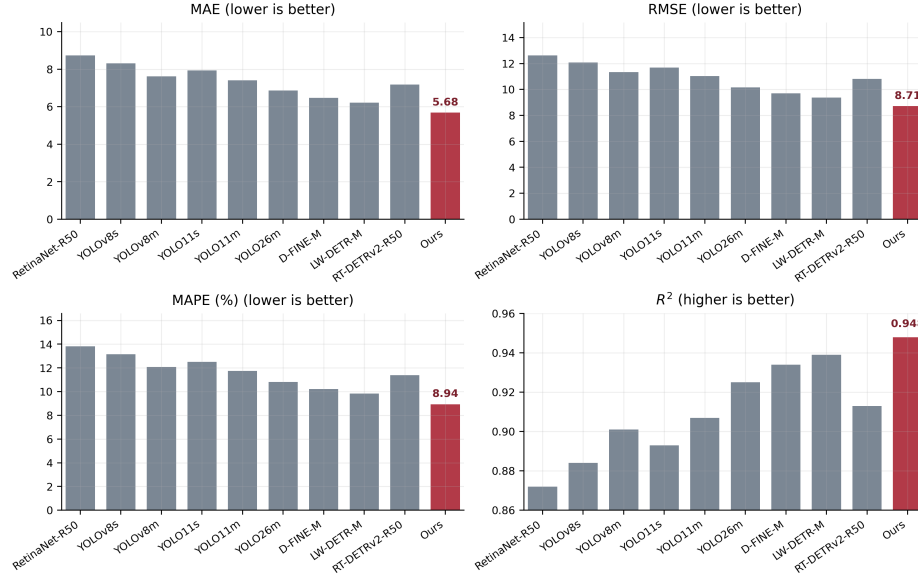


Figure 6: Image-level counting comparison on the GWHD2021 validation set using the data from Table 5. Lower MAE, RMSE, and MAPE indicate smaller count errors, whereas higher R^2 indicates stronger agreement between predicted and annotated image-level counts.

Table 6: Pathway ablation study on the GWHD2021 validation set using RT-DETRv2-R50 as the controlled reference. AP_S denotes COCO small-object AP. All variants were trained and evaluated under the same protocol.

Variant	Context-Detail	Detail Branch	Compact Query	AP	AP50	AP75	AP-S	Delta AP	Params (M)	GFLOPs	FPS
Controlled reference	No	No	No	50.68	91.74	56.92	47.96	—	42.07	171.6	108.46
Context-detail representation	Yes	No	No	51.41	92.08	57.92	48.94	+0.73	42.54	174.1	106.91
Detail branch emphasis	No	Yes	No	51.03	91.92	57.34	48.58	+0.35	42.24	172.5	107.68
Compact-query allocation	No	No	Yes	51.37	92.01	57.76	49.23	+0.69	42.07	171.6	107.12
Context-detail + Detail branch	Yes	Yes	No	52.05	92.32	58.60	49.82	+1.37	42.71	175.0	105.84
Context-detail + Compact query	Yes	No	Yes	52.56	92.58	59.18	50.64	+1.88	42.54	174.1	105.19
Detail branch + Compact query	No	Yes	Yes	52.12	92.45	58.82	50.12	+1.44	42.24	172.5	106.25
TSECAF-DETR-R50	Yes	Yes	Yes	53.62	93.18	60.41	51.46	+2.94	42.81	176.3	102.80

strict localization and small-object behavior are especially relevant to dense wheat scenes, where adjacent heads may be separated by narrow visual gaps.

Table 6 supports three observations. First, each partial pathway intervention improved the controlled reference, indicating that both representation formation and query allocation are active points in the dense-head failure chain. Among the partial variants, context-detail representation produced the largest AP gain, suggesting that semantic-detail interaction is a major factor in dense wheat scenes. This agrees with the visual structure of the task: reliable separation of adjacent heads depends on preserving local detail

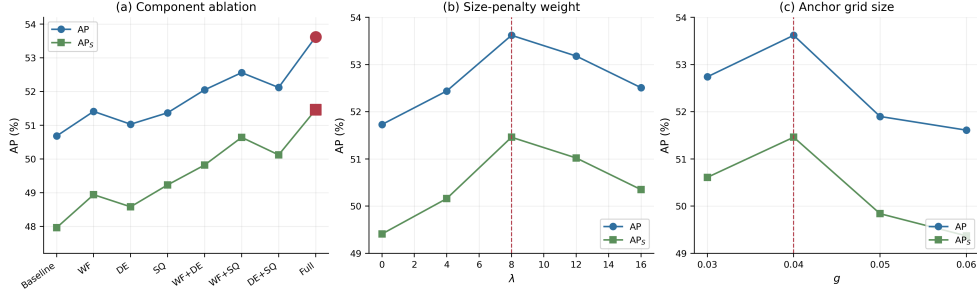


Figure 7: Visual summary of the pathway ablation and compact-query hyperparameter sensitivity results from Tables 6 and 7. TSECAF-DETR-R50 obtains the strongest AP and AP_S, and the selected query settings $\lambda = 8.0$ and $g = 0.04$ correspond to the best observed operating point.

while maintaining canopy-level context.

Second, compact-query allocation improved AP_S more strongly than the detail-branch-only variant. This agrees with the intended role of *agnostic_small*, which reallocates limited decoder queries toward compact wheat-head proposals that might otherwise be underrepresented during refinement. Third, the jointly active pathway outperformed the corresponding partial variants, and TSECAF-DETR-R50 achieved the best values across all reported metrics. The full architecture improved AP by 2.94 points and AP_S by 3.50 points relative to the controlled reference while adding only 0.74 M parameters and 4.7 GFLOPs, with a throughput reduction of 5.66 FPS. This pattern is not expected from uncoordinated additions. The representation variants change the candidate evidence available to the encoder proposal heads, and compact-query allocation changes how decoder capacity is assigned to that evidence. Their strongest performance when activated together supports the proposed evidence-to-query pathway interpretation.

To examine whether the compact-query gain depends strongly on hyperparameter choice, a sensitivity analysis was performed for the size-penalty weight λ and the anchor grid size g , as shown in Table 7. These two hyperparameters control different aspects of the proposal-to-query allocation process. The parameter λ determines how strongly large predicted boxes are penalized during top- K ranking, whereas g controls the geometric scale of the initial anchor prior before decoder refinement. Because wheat images contain compact targets and natural variation in head size and viewing distance, the selected setting should be effective without being narrowly tuned.

Table 7: Sensitivity analysis of the compact-query allocation hyperparameters on the GWHD2021 validation set using RT-DETRv2-R50. Panel A varies the size-penalty weight λ with fixed anchor grid size $g = 0.04$. Panel B varies g with fixed $\lambda = 8.0$.

A. Sensitivity to the size-penalty weight lambda (g=0.04)						
lambda	AP	AP50	AP75	AP-S	Delta AP vs. lambda=0	FPS
0	51.73	92.16	58.21	49.41	–	102.93
4	52.44	92.71	59.08	50.16	+0.71	102.89
8	53.62	93.18	60.41	51.46	+1.89	102.80
12	53.18	92.96	59.92	51.02	+1.45	102.76
16	52.51	92.48	59.13	50.35	+0.78	102.83
B. Sensitivity to the anchor grid size g (lambda=8.0)						
g	AP	AP50	AP75	AP-S	Delta AP vs. g=0.05	FPS
0.03	52.74	92.77	59.46	50.61	+0.84	102.86
0.04	53.62	93.18	60.41	51.46	+1.72	102.80
0.05	51.90	92.25	58.43	49.84	–	102.91
0.06	51.61	92.06	58.18	49.37	-0.29	102.88

The sensitivity results indicate that compact-query allocation is not tied to a narrow parameter setting. In Panel A of Table 7, performance improved as λ increased from 0 to 8, after which the gain saturated and then declined slightly. This trend follows the expected behavior of the ranking rule. When λ is too small, selection remains close to objectness-dominated ranking and does not sufficiently emphasize compact heads. When λ becomes too large, the ranking begins to over-penalize larger but valid wheat-head proposals. A moderate value of $\lambda = 8.0$ provided the best balance, as also shown in Figure 7. Panel B shows a similar tendency for the anchor grid size. Compared with the standard prior of $g = 0.05$, reducing the prior to $g = 0.04$ improved AP and AP_S, indicating that a slightly smaller initialization is beneficial for dense wheat scenes. Shrinking the prior further to $g = 0.03$ did not provide additional benefit, suggesting that overly small anchors may become too conservative when head size varies across images. These results support the final hyperparameter choice used in the main experiments and indicate that the observed gain is robust within a reasonable parameter range.

6. Discussion

6.1. Interpretation of the Detection Gains

The results support the representation-query bottleneck view introduced in this paper. At the feature level, adjacent heads are often separated by narrow gaps, weak boundaries, awns, or subtle texture differences. If high-resolution detail is weakened during feature aggregation, nearby heads may be merged into a single response, and small partially occluded heads may disappear before proposal generation. At that point the query selector cannot recover the lost instance because the candidate never enters the proposal set as a distinct hypothesis. The ablation results support this interpretation. Context-detail representation produced the largest partial AP gain, and TSECAF-DETR-R50 further improved AP75 and AP_S. These metrics are particularly relevant to dense wheat scenes because they measure strict localization and small-object behavior, which most directly expose representation dilution.

At the query level, proposal ranking acts as the second bottleneck in the same pathway. RT-DETRv2 initializes decoder queries from encoder-side candidates, so valid wheat-head regions must survive the top- K selection stage before they can be refined by the decoder. In single-class wheat head detection, this competition is mainly geometric rather than semantic: candidate regions should correspond to individual heads rather than broad canopy structures. Compact-query allocation improved AP_S, indicating that compact proposals benefit from receiving more decoder capacity. The best performance was obtained when context-detail representation and compact-query allocation were jointly active. This pattern is important because it indicates co-dependence rather than additive stacking: the representation unit makes compact evidence explicit enough to become proposals, and the allocation rule then protects that evidence during decoder initialization. Either side alone addresses only half of the failure chain.

A practical advantage of TSECAF-DETR is its architectural economy. The detector does not increase complexity uniformly; instead, it concentrates modeling capacity at the two points where the same compact candidates are most likely to be lost: semantic-detail representation before proposal generation and compact-object query allocation before decoder refinement. Across R18, R34, and R50, the TSECAF-DETR variants improved accuracy while preserving a clear accuracy-efficiency trade-off. YOLO-family models remained faster and lighter, but TSECAF-DETR-R50 provided stronger AP,

AP75, AP_S, and recall than the standard YOLO baselines while still running at more than 100 FPS on the evaluation GPU. The additional comparison with YOLO26, D-FINE, and LW-DETR shows that TSECAF-DETR-R50 achieved the strongest AP, AP75, and AP_S among the tested recent baselines while remaining faster than the medium-size D-FINE and LW-DETR variants. Together with the backbone-scale results and compact-query hyperparameter sensitivity analysis, these findings suggest that the improvement is consistent across model capacities and parameter settings.

6.2. Image-Level Effects and Cross-Dataset Generalization

Detection accuracy is most informative when it is connected to the behavior of localized instances. The counting experiment provides this link without introducing a separate counting model. Compared with RT-DETRv2-R50, TSECAF-DETR-R50 reduced MAE from 7.18 to 5.68 heads per image and improved R^2 from 0.913 to 0.948. This pattern is consistent with the intended role of the method in dense small-object scenes, where missed and merged detections can accumulate into count error. It also gives an interpretable consequence of the representation-query design: the method does not merely raise confidence scores, but increases the chance that individual compact heads are both represented as separable candidates and refined as decoder queries. This reduces localization misses and duplicate/merged detections that would otherwise distort image-level counts.

The transfer results further indicate that the proposed design is not limited to the GWHD2021 validation split. The GWHD2021-trained model retained consistent gains on the Wheat Ears Detection Dataset and on labeled subsets of Global Wheat CodaLab without target-domain fine-tuning. This does not remove the need for domain adaptation or threshold calibration when image statistics change substantially, but it suggests that the coupled evidence-to-query pathway improves robustness under realistic distribution shifts [61, 62]. In this sense, public wheat datasets serve as reproducible dense small-object benchmarks: they provide controlled tests of localization, transfer, count consistency, and throughput without requiring claims about a complete application pipeline.

The backbone-scale results provide practical operating choices. TSECAF-DETR-R18 favors throughput, TSECAF-DETR-R50 favors localization accuracy, and TSECAF-DETR-R34 offers an intermediate operating point. This flexibility is useful because dense-image workloads differ in computational constraints: large image archives may require fast batch processing,

whereas strict localization analysis may prioritize AP75, AP_S, and lower count error.

6.3. Limitations and Future Work

Several limitations should be considered. First, this article focuses on single-class wheat head detection using bounding boxes. This formulation is suitable for benchmark comparison and image-level counting, but it does not provide more detailed structural information such as ear segmentation, spikelet-level traits, or multi-organ relationships. Second, although stable gains were observed across three wheat datasets, the compact-query hyperparameters were selected for dense wheat scenes and may require adjustment under substantially different imaging scales, target morphologies, or sensor settings. The sensitivity analysis indicates that the method is not narrowly tuned within the tested range, but it does not guarantee optimal behavior under all visual domains.

Third, the experiments used public datasets rather than a newly collected deployment dataset with repeated acquisition, plot metadata, or independent manual count records. Therefore, the study validates the detector as a transferable dense-recognition design, but it does not by itself validate a complete agronomic measurement pipeline. Fourth, FPS was measured in a controlled single-GPU environment rather than on embedded hardware. The present results should therefore be interpreted as an improvement in dense small-object detection and image-level count consistency, not as an end-to-end deployment study.

Future work should evaluate broader generalization across additional image sources, growth stages, viewpoints, resolutions, and image qualities; combine the detector with domain adaptation or self-supervised pretraining; test lightweight variants on edge hardware; and extend the method toward richer cereal-organ analysis, such as ear segmentation or multi-organ detection. These extensions would clarify how far context-detail representation and compact-query allocation can transfer beyond benchmark wheat head detection.

7. Conclusion

This article presented TSECAF-DETR, a real-time transformer detector for dense wheat head detection. The central contribution is a representation-query co-design that treats dense small-object detection as an evidence-to-

query allocation pathway. Dense small targets are commonly lost first during semantic-detail fusion before proposal generation and then during proposal-to-query allocation before decoder refinement. Wheat-TS-ECAF strengthens the interaction between semantic context and local head-detail cues so that compact head hypotheses remain explicit, while the area-aware proposal ranking strategy increases the likelihood that those hypotheses enter decoder refinement.

TSECAF-DETR improved detection accuracy, transfer behavior, and image-level counting performance. On GWHD2021, TSECAF-DETR-R50 achieved 53.62 AP, 93.18 AP₅₀, 60.41 AP₇₅, and 51.46 AP_S while maintaining 102.80 FPS. It outperformed representative Faster R-CNN, RetinaNet, YOLO, RT-DETRv2, YOLO26, D-FINE, and LW-DETR baselines in the source-domain comparison. Without target-domain fine-tuning, the same GWHD2021-trained detector improved direct transfer performance on the Wheat Ears Detection Dataset and the labeled subsets of Global Wheat Co-daLab. In the counting analysis, it reduced image-level MAE from 7.18 to 5.68 relative to RT-DETRv2-R50.

Ablation and sensitivity analyses showed that representation formation and query allocation are co-dependent and that the gain is stable within a reasonable hyperparameter range. Overall, the results indicate that dense small-object detection benefits from treating visual evidence preservation and decoder capacity allocation as a unified design problem. TSECAF-DETR therefore provides a benchmarked design principle for aligning compact-evidence formation with decoder-query allocation, yielding stronger localized instances and more consistent image-level counts without sacrificing practical inference efficiency.

CRedit authorship contribution statement

Conceptualization, Z.W. and J.L.; methodology, Z.W. and J.L.; software, Z.W.; validation, Z.W. and Z.Z.; formal analysis, Z.W. and J.L.; investigation, Z.W. and Z.Z.; data curation, Z.W. and L.W.; writing—original draft preparation, Z.W.; writing—review and editing, J.L., Z.Z., L.W. and N.Y.; visualization, Z.W.; supervision, J.L.; project administration, J.L. and N.Y. All authors have read and agreed to the published version of the manuscript.

Funding

This research received no external funding.

Ethics Statement

Institutional Review Board Statement: Not applicable. Informed Consent Statement: Not applicable.

Data Availability Statement

All datasets used in this article are publicly available. The GWHD2021 dataset, the Wheat Ears Detection Dataset, and the Global Wheat Head Dataset 2020 Challenge version (Global Wheat CodaLab release) can be accessed from the sources cited in the References. The project code repository containing the implementation, configuration files, and evaluation scripts is publicly available at <https://github.com/Noon-Knowledge/TSECAF-DETR>.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI ChatGPT and Codex to improve language clarity and manuscript organization. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Abbreviations

The following abbreviations are used in this manuscript:

AP	Average precision
D-FINE	Fine-grained distribution refinement detector
FPN	Feature pyramid network
GWHD	Global Wheat Head Detection
LW-DETR	Lightweight Detection Transformer
NMS	Non-maximum suppression
PR	Precision-recall
RT-DETR	Real-Time Detection Transformer

References

- [1] R. T. Furbank and M. Tester, "Phenomics—technologies to relieve the phenotyping bottleneck," *Trends Plant Sci.*, vol. 16, no. 12, pp. 635–644, 2011, doi: 10.1016/j.tplants.2011.09.005.
- [2] A. Walter, F. Liebisch, and A. Hund, "Plant phenotyping: From bean weighing to image analysis," *Plant Methods*, vol. 11, Art. no. 14, 2015, doi: 10.1186/s13007-015-0056-8.
- [3] J. R. Ubbens and I. Stavness, "Deep plant phenomics: A deep learning platform for complex plant phenotyping tasks," *Front. Plant Sci.*, vol. 8, Art. no. 1190, 2017, doi: 10.3389/fpls.2017.01190.
- [4] F. Perez-Sanz, P. J. Navarro, and M. Egea-Cortines, "Plant phenomics: An overview of image acquisition technologies and image data analysis algorithms," *GigaScience*, vol. 6, no. 11, Art. no. gix092, 2017, doi: 10.1093/gigascience/gix092.
- [5] E. David *et al.*, "Global Wheat Head Detection (GWHD) dataset: A large and diverse dataset of high-resolution RGB-labelled images to develop and benchmark wheat head detection methods," *Plant Phenomics*, vol. 2020, Art. no. 3521852, 2020, doi: 10.34133/2020/3521852.
- [6] E. David *et al.*, "Global Wheat Head Detection 2021: An improved dataset for benchmarking wheat head detection methods," *Plant Phenomics*, vol. 2021, Art. no. 9846158, 2021, doi: 10.34133/2021/9846158.
- [7] S. Madec, X. Jin, H. Lu, B. de Solan, S. Liu, F. Duyme, E. Heritier, and F. Baret, "Ear density estimation from high resolution RGB imagery using deep learning technique," *Agric. Forest Meteorol.*, vol. 264, pp. 225–234, 2019, doi: 10.1016/j.agrformet.2018.10.013.
- [8] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 2117–2125, doi: 10.1109/CVPR.2017.106.
- [9] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 8759–8768, doi: 10.1109/CVPR.2018.00913.

- [10] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 10781–10790, doi: 10.1109/CVPR42600.2020.01079.
- [11] Y. Li, Y. Chen, N. Wang, and Z. Zhang, "Scale-aware trident networks for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 6054–6063, doi: 10.1109/ICCV.2019.00615.
- [12] C. Guo, B. Fan, Q. Zhang, S. Xiang, and C. Pan, "AugFPN: Improving multi-scale feature learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 12595–12604, doi: 10.1109/CVPR42600.2020.01261.
- [13] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Computer Vision–ECCV 2020*. Cham, Switzerland: Springer, 2020, pp. 213–229, doi: 10.1007/978-3-030-58452-8_13.
- [14] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021. [Online]. Available: <https://arxiv.org/abs/2010.04159>.
- [15] Y. Zhao *et al.*, "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 16965–16974, doi: 10.1109/CVPR52733.2024.01605.
- [16] W. Lv, Y. Zhao, Q. Chang, K. Huang, G. Wang, and Y. Liu, "RT-DETRv2: Improved baseline with bag-of-freebies for real-time detection transformer," *arXiv*, 2024, arXiv:2407.17140, doi: 10.48550/arXiv.2407.17140.
- [17] J. L. Araus and J. E. Cairns, "Field high-throughput phenotyping: The new crop breeding frontier," *Trends Plant Sci.*, vol. 19, no. 1, pp. 52–61, 2014, doi: 10.1016/j.tplants.2013.09.008.
- [18] J. A. Fernandez-Gallego, S. C. Kefauver, N. Aparicio Gutierrez, M. T. Nieto-Taladriz, and J. L. Araus, "Wheat ear counting in-field conditions: High throughput and low-cost approach using RGB images,"

- Plant Methods*, vol. 14, Art. no. 22, 2018, doi: 10.1186/s13007-018-0289-4.
- [19] A. Kamilaris and F. X. Prenafeta-Boldu, "Deep learning in agriculture: A survey," *Comput. Electron. Agric.*, vol. 147, pp. 70–90, 2018, doi: 10.1016/j.compag.2018.02.016.
 - [20] K. G. Liakos, P. Busato, D. Moshou, S. Pearson, and D. Bochtis, "Machine learning in agriculture: A review," *Sensors*, vol. 18, no. 8, Art. no. 2674, 2018, doi: 10.3390/s18082674.
 - [21] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, 2017, doi: 10.1109/TPAMI.2016.2577031.
 - [22] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Computer Vision—ECCV 2016*. Cham, Switzerland: Springer, 2016, pp. 21–37, doi: 10.1007/978-3-319-46448-0_2.
 - [23] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2980–2988, doi: 10.1109/ICCV.2017.324.
 - [24] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," *arXiv*, 2020, arXiv:2004.10934. [Online]. Available: <https://arxiv.org/abs/2004.10934>.
 - [25] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLO," software, version 8.0.0, 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>. Accessed: May 17, 2026.
 - [26] G. Jocher and J. Qiu, "Ultralytics YOLO11," software, version 11.0.0, 2024. [Online]. Available: <https://github.com/ultralytics/ultralytics>; <https://docs.ultralytics.com/models/yolo11/>. Accessed: May 17, 2026.
 - [27] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," *arXiv*, 2025, arXiv:2502.12524, doi: 10.48550/arXiv.2502.12524.

- [28] Ultralytics, "Ultralytics YOLO26," software, 2026. [Online]. Available: <https://docs.ultralytics.com/models/yolo26/>. Accessed: May 21, 2026.
- [29] I. Sa, Z. Ge, F. Dayoub, B. Upcroft, T. Perez, and C. McCool, "Deep-Fruits: A fruit detection system using deep neural networks," *Sensors*, vol. 16, no. 8, Art. no. 1222, 2016, doi: 10.3390/s16081222.
- [30] S. Bargoti and J. Underwood, "Deep fruit detection in orchards," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2017, pp. 3626–3633, doi: 10.1109/ICRA.2017.7989417.
- [31] M. P. Pound, J. A. Atkinson, D. M. Wells, T. P. Pridmore, and A. P. French, "Deep learning for multi-task plant phenotyping," in *Proc. IEEE Int. Conf. Comput. Vis. Workshops (ICCVW)*, 2017, pp. 2055–2063.
- [32] M. M. Hasan, J. P. Chopin, H. Laga, and S. J. Miklavcic, "Detection and analysis of wheat spikes using convolutional neural networks," *Plant Methods*, vol. 14, Art. no. 100, 2018.
- [33] S. Madec, "Wheat Ears Detection Dataset," Zenodo, 2018, doi: 10.5281/zenodo.1471626. Accessed: May 17, 2026.
- [34] B. Gong, D. Ergu, Y. Cai, and B. Ma, "Real-time detection for wheat head applying deep neural network," *Sensors*, vol. 21, no. 1, Art. no. 191, 2021, doi: 10.3390/s21010191.
- [35] S. Bhagat, M. Kokare, R. Haswani, P. Hambarde, and R. Kamble, "WheatNet-Lite: A novel light weight network for wheat head detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, 2021, pp. 1332–1341, doi: 10.1109/ICCVW54120.2021.00154.
- [36] S. Khaki, N. Safaei, H. Pham, and L. Wang, "WheatNet: A lightweight convolutional neural network for high-throughput image-based wheat head detection and counting," *Neurocomputing*, vol. 489, pp. 78–89, 2022, doi: 10.1016/j.neucom.2022.03.017.
- [37] J. Sun *et al.*, "Wheat head counting in the wild by an augmented feature pyramid networks-based convolutional neural network," *Comput. Electron. Agric.*, vol. 193, Art. no. 106705, 2022, doi: 10.1016/j.compag.2022.106705.

- [38] Y. Zhang, M. Li, X. Ma, X. Wu, and Y. Wang, "High-precision wheat head detection model based on one-stage network and GAN model," *Front. Plant Sci.*, vol. 13, Art. no. 787852, 2022, doi: 10.3389/fpls.2022.787852.
- [39] P. Sadeghi-Tehran, N. Virlet, E. M. Ampe, P. Reyns, and M. J. Hawkesford, "DeepCount: In-field automatic quantification of wheat spikes using simple linear iterative clustering and deep convolutional neural networks," *Front. Plant Sci.*, vol. 10, Art. no. 1176, 2019, doi: 10.3389/fpls.2019.01176.
- [40] K. Najafian, A. Ghanbari, I. Stavness, L. Jin, G. H. Shirdel, and F. Maleki, "A semi-self-supervised learning approach for wheat head detection using extremely small number of labeled samples," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, 2021, pp. 1342–1351, doi: 10.1109/ICCVW54120.2021.00155.
- [41] J. Zhu *et al.*, "Detecting wheat heads from UAV low-altitude remote sensing images using deep learning based on transformer," *Remote Sens.*, vol. 14, no. 20, Art. no. 5141, 2022, doi: 10.3390/rs14205141.
- [42] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 7132–7141, doi: 10.1109/CVPR.2018.00745.
- [43] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19, doi: 10.1007/978-3-030-01234-2_1.
- [44] D. Meng *et al.*, "Conditional DETR for fast training convergence," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 3651–3660, doi: 10.1109/ICCV48922.2021.00363.
- [45] P. Gao, M. Zheng, X. Wang, J. Dai, and H. Li, "Fast convergence of DETR with spatially modulated co-attention," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 3621–3630, doi: 10.1109/ICCV48922.2021.00360.
- [46] Y. Wang, X. Zhang, T. Yang, and J. Sun, "Anchor DETR: Query design for transformer-based detector," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 3, 2022, pp. 2567–2575, doi: 10.1609/aaai.v36i3.20158.

- [47] S. Liu *et al.*, "DAB-DETR: Dynamic anchor boxes are better queries for DETR," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022. [Online]. Available: <https://arxiv.org/abs/2201.12329>.
- [48] P. Sun *et al.*, "Sparse R-CNN: End-to-end object detection with learnable proposals," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 14454–14463, doi: 10.1109/CVPR46437.2021.01422.
- [49] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, "DN-DETR: Accelerate DETR training by introducing query denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 13619–13627, doi: 10.1109/CVPR52688.2022.01325.
- [50] H. Zhang *et al.*, "DINO: DETR with improved denoising anchor boxes for end-to-end object detection," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023. [Online]. Available: <https://arxiv.org/abs/2203.03605>.
- [51] Y. Peng, H. Li, P. Wu, Y. Zhang, X. Sun, and F. Wu, "D-FINE: Redefine regression task in DETRs as fine-grained distribution refinement," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025, doi: 10.48550/arXiv.2410.13842.
- [52] Q. Chen *et al.*, "LW-DETR: A transformer replacement to YOLO for real-time detection," *arXiv*, 2024, arXiv:2406.03459, doi: 10.48550/arXiv.2406.03459.
- [53] E. David, "Global Wheat Head Dataset – 2020 Challenge version," Zenodo, 2020, doi: 10.5281/zenodo.4298502. Accessed: May 17, 2026.
- [54] T.-Y. Lin *et al.*, "Microsoft COCO: Common objects in context," in *Computer Vision–ECCV 2014*. Cham, Switzerland: Springer, 2014, pp. 740–755, doi: 10.1007/978-3-319-10602-1_48.
- [55] PyTorch, "Models and pre-trained weights," TorchVision documentation, 2026. [Online]. Available: <https://docs.pytorch.org/vision/stable/models.html>. Accessed: May 22, 2026.

- [56] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019. [Online]. Available: <https://arxiv.org/abs/1711.05101>.
- [57] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.
- [58] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017. [Online]. Available: <https://arxiv.org/abs/1608.03983>.
- [59] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018. [Online]. Available: <https://arxiv.org/abs/1710.09412>.
- [60] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 658–666, doi: 10.1109/CVPR.2019.00075.
- [61] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, 2010, doi: 10.1109/TKDE.2009.191.
- [62] M. Wang and W. Deng, "Deep visual domain adaptation: A survey," *Neurocomputing*, vol. 312, pp. 135–153, 2018, doi: 10.1016/j.neucom.2018.05.083.